

FedFraudShield: A Federated Learning Framework for Privacy-Preserving Fraud Detection Across Microfinance Institutions in Uganda

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Abstract

Internal fraud costs Uganda's 28,500+ Savings and Credit Cooperative Organizations (SACCOs) and microfinance institutions (MFIs) an estimated UGX 100–250 billion annually, yet no dedicated AI-powered fraud detection solution exists for this sector. A key barrier is data scarcity: individual institutions hold limited transaction histories, and privacy regulations under Uganda's Data Protection and Privacy Act (DPPA 2019) restrict raw data sharing. We propose **FedFraudShield**, a federated learning framework that enables multiple SACCOs and MFIs to collaboratively train fraud detection models without sharing raw data. Our framework employs a hub-and-spoke architecture with horizontal federated learning, differential privacy, and communication-efficient design for low-bandwidth environments. We present experiments using the PaySim dataset (modeled on African mobile money transactions), partitioned across 15 simulated institutions with non-IID distributions. Federated models achieve AUPRC of 0.945 compared to 0.643 for locally-trained models—a **47% relative improvement**—while institutions with zero fraud history gain full detection capability through federation. Differential privacy at $\epsilon=5.0$ maintains performance above the local-only baseline (AUPRC 0.692 vs 0.643). Total communication cost of 30 MB over 20 rounds (100 KB per node per round) makes deployment feasible on 3G connections. This work-in-progress establishes the feasibility and architectural design of privacy-preserving collaborative fraud detection for Uganda's microfinance sector.

Keywords: Federated Learning, Fraud Detection, Microfinance, SACCOs, Privacy-Preserving ML, Uganda, DPPA 2019

1. Introduction

Uganda's cooperative financial sector, comprising over 28,556 registered SACCOs and 150+ licensed microfinance institutions (MFIs), manages more than UGX 5 trillion in assets and serves approximately 18 million members [1]. Internal fraud—including ghost loans, collusion rings, officer self-lending, and payment manipulation—erodes 2–5% of portfolio value annually, translating to estimated losses of UGX 100–250 billion (\$27–68M USD) [2]. The PROFIRA assessment found that of 453 monitored SACCOs, 312 were struggling due to fraud and poor governance, and 64 had collapsed entirely [3].

Despite the severity of this problem, applying machine learning to fraud detection in Uganda's SACCO sector faces a fundamental data challenge. **FraudShield Uganda**, an AI-powered fraud detection project initiated by the author, encountered this barrier directly: individual SACCOs hold limited transaction histories (often only 6–24 months of digital records), fraud labels are sparse and inconsistent, and institutions are reluctant to share sensitive financial data with third parties. A single SACCO with 500 members may process only 5,000–10,000 transactions per year, with fewer than 50 confirmed fraud cases—far too few to train robust ML models. This experience motivated the present work: if no single institution has sufficient data, could multiple institutions collaborate to build better fraud detection models without compromising data privacy?

Federated learning (FL) [4] offers a compelling answer. In FL, a shared model is trained across decentralized data sources by exchanging only model parameters—never raw data. This approach is particularly suited to Uganda's regulatory environment, where the DPPA 2019 restricts personal data processing but provides explicit exceptions for fraud prevention (Section 7(2)(b)(iii)) [5]. Furthermore, the 2022 Credit Reference Bureau Regulations [6] established a precedent for cross-institutional financial data sharing, suggesting regulatory receptivity to collaborative approaches.

We propose **FedFraudShield**, a federated learning framework specifically designed for fraud detection across Uganda's SACCOs and MFIs. Our contributions are: (1) a hub-and-spoke FL architecture tailored to the

technical and regulatory constraints of Ugandan microfinance, (2) a privacy analysis demonstrating compliance with DPPA 2019, and (3) experimental results demonstrating that federated training achieves 47% higher AUPRC than local-only models, with institutions lacking fraud history gaining detection capability through federation.

2. Related Work

Federated learning for fraud detection. Yang et al. [7] proposed FFD, the first FL-based credit card fraud detection system, achieving AUC of 95.5% across simulated banks. Subsequent work integrated meta-learning [8], graph neural networks [9], and reinforcement learning [10] into federated fraud detection. Recent work by Roth et al. [11] demonstrated that FedAvg, FedProx, and FedOpt achieve comparable F1 scores (~ 0.903) for payment fraud detection, with convergence within 3–5 rounds. Nguyen et al. [12] proposed FL FraDet for mobile money fraud detection, demonstrating that performance-aware aggregation outperforms FedAvg by 10.1% under non-IID conditions.

African financial fraud detection. Azamuke et al. [13, 14] at Makerere University developed MoMTSim, a synthetic mobile money transaction simulator calibrated on Sub-Saharan African data. Ampumuza et al. [15] conducted a systematic review of AI-driven fraud detection in SACCO transactions, identifying the need for hybrid models and privacy-preserving analytics. Ojino and Ndolo [16] applied knowledge graphs to SACCO fraud detection in Kenya. However, **no prior work has applied federated learning to SACCO or MFI fraud detection in Africa**—a gap this paper addresses.

Privacy-preserving FL in finance. Zhang et al. [17] proposed HyFL, a hybrid FL framework for financial crime detection that won the PETs Prize Challenge. DPFedBank [18] and Fed-RD [19] formalized privacy guarantees for financial FL using differential privacy and secure multi-party computation. Fabila et al. [20] demonstrated FL across five African countries using low-resource edge devices.

3. Proposed Framework: FedFraudShield

3.1 System Architecture

FedFraudShield employs a **hub-and-spoke architecture** consisting of four layers (Table 1): (i) **Local Training Nodes** deployed at each participating SACCO/MFI, (ii) a **Privacy Layer** for differential privacy and gradient compression, (iii) a **Central Aggregation Server** hosted within Uganda for DPPA 2019 compliance, and (iv) an **Alert Layer** for real-time fraud scoring.

Layer	Component	Function
Local Node (per institution)	Data Preprocessor + Local Trainer	Schema mapping, feature engineering, local model training on private data
Privacy Layer	DP Module + Gradient Compressor	Differential privacy noise injection, quantization for bandwidth efficiency
Aggregation Server	Secure Aggregator + Model Registry	Weighted federated averaging, global model versioning and distribution
Alert Layer	Inference Engine + Alert Dispatcher	Real-time scoring with global model, SMS/dashboard alerts to SACCO staff

Table 1. FedFraudShield architecture layers and components.

3.2 Federated Learning Protocol

We adopt **horizontal federated learning** (HFL) as SACCOs share similar transaction schemas (deposits, withdrawals, loan disbursements, repayments) but hold different member populations. The training protocol proceeds in rounds: **Round initialization:** The aggregation server initializes a global model G_0 (a lightweight MLP with 128–64–32 hidden units) and distributes it to all K participating institutions. **Local training:** Each institution k trains on its private dataset D_k for E local epochs, producing updated parameters w_k . To handle class imbalance (fraud typically $<1\%$ of transactions), each node applies local SMOTE oversampling and weighted loss functions. **Privacy-preserving upload:** Each node clips gradients to bound L2 sensitivity and adds calibrated Gaussian noise for (ϵ, δ) -differential privacy. **Aggregation:** The server computes the weighted average: $w_{t+1} = \sum_k (n_k/n) \cdot w_k$, where n_k is the number of samples at institution k . We evaluate both FedAvg [4]

and FedProx [21], which adds a proximal term to handle heterogeneity. **Distribution:** The updated global model is broadcast to all nodes for the next round.

3.3 Handling Uganda-Specific Challenges

Non-IID data distributions: Different SACCOs experience different fraud patterns. Rural SACCOs may see more ghost loans, while urban MFIs encounter more sophisticated collusion rings. We address this through FedProx's proximal regularization and personalization layers that adapt the global model to local fraud patterns.

Schema heterogeneity: SACCOs use different software systems (Ensibuuko, Kanza Banking, Excel/QuickBooks) with varying column names and data formats. The local node includes a schema mapping module that normalizes institution-specific data to a common feature schema comprising 17 core transaction features. **Low bandwidth:** Rural Uganda has variable internet connectivity. Our model requires only ~100 KB per node per round, feasible even on 3G mobile connections.

3.4 DPPA 2019 Compliance

FedFraudShield complies with Uganda's DPPA 2019 [5] through three mechanisms. First, **no raw data leaves any institution**—only model parameter updates are transmitted, satisfying the data minimality principle (Section 14). Second, **differential privacy** ensures that model updates cannot be reverse-engineered to reveal individual transactions. Third, the fraud prevention exception under Section 7(2)(b)(iii) provides explicit legal basis for processing personal data for the prevention and detection of offences. The aggregation server is hosted within Uganda, avoiding cross-border transfer restrictions (Section 19). The 2022 CRB Regulations [6] further establish precedent for cross-institutional financial data sharing in Uganda.

4. Experiments

4.1 Experimental Setup

Dataset: We use a stratified 15% subsample of the PaySim dataset [23], a synthetic financial dataset generated from real mobile money transaction logs from an African country. The subsample preserves all 8,213 fraud cases across 954,393 transactions (fraud rate 0.86%), providing sufficient signal while enabling tractable computation. PaySim contains five transaction types (CASH_IN, CASH_OUT, DEBIT, PAYMENT, TRANSFER), closely mirroring the patterns observed in Ugandan mobile money and SACCO transactions.

Simulating heterogeneous institutions: We partition the data across $K=15$ simulated institutions using Dirichlet distribution sampling with concentration parameter $\alpha=0.5$ to create non-IID label distributions [24]. Quantity skew follows a log-normal distribution reflecting real variance in Uganda's SACCO sector—institution sizes range from 18,619 to 166,604 transactions. Five institutions receive no fraud cases at all, simulating SACCOs with no labeled fraud history. **Model:** A lightweight MLP (128–64–32 hidden units, ~14,000 parameters) with BatchNorm, ReLU activations, and dropout. We engineer 17 features including transaction amount, type, balance deltas, time-of-day, and account balance ratios. **Baselines:** (1) **Local-only**—each institution trains solely on its own data; (2) **FedAvg** [4]; (3) **FedProx** [21] ($\mu=0.01$); (4) **Centralized**—upper-bound oracle with all data pooled. We apply (ϵ, δ) -DP with $\epsilon \in \{1.0, 5.0\}$. All experiments use 20 federated rounds, 5 local epochs, and the Flower-compatible PyTorch framework [25]. Primary metric: AUPRC (preferred for class imbalance).

4.2 Results

Method	AUPRC	AUROC	F1	Recall	Comm.
Local-only (avg)	0.643	0.829	0.427	0.768	0 MB
Local-only (best)	0.959	0.999	0.758	1.000	0 MB
Local-only (worst)	0.005	0.192	0.006	0.033	0 MB
FedAvg (20 rounds)	0.945	0.996	0.090	0.991	30 MB
FedProx (20 rounds)	0.890	0.995	0.207	0.989	30 MB
FedProx + DP ($\epsilon=5$)	0.692	0.972	0.475	0.312	30 MB
FedProx + DP ($\epsilon=1$)	0.079	0.743	0.190	0.227	30 MB
Centralized (oracle)	0.977	1.000	0.667	0.995	N/A

Table 2. Experimental results on PaySim (15% stratified subsample, all fraud cases preserved) partitioned across 15 simulated institutions ($\alpha=0.5$ Dirichlet non-IID). Best federated method highlighted. DP = Differential Privacy.

4.3 Key Findings

Federation dramatically outperforms local training. FedAvg achieves AUPRC of 0.945, a **47% relative improvement** over the local-only average of 0.643. Critically, five institutions with zero local fraud cases—which cannot train any fraud detector independently—gain full detection capability through federation (AUPRC = 0.0 locally vs. benefiting from the shared global model). This demonstrates the framework's transformative value for small or newly-digital SACCOs.

Federated models achieve near-perfect recall. Both FedAvg and FedProx achieve recall above 0.98, meaning they catch over 98% of fraudulent transactions. The lower F1 scores (0.090–0.207) reflect a precision trade-off: the models flag more false positives, which is an acceptable trade-off in fraud detection where missing a real fraud case is far more costly than investigating a false alarm. Precision can be improved through threshold tuning at deployment.

FedAvg outperforms FedProx on AUPRC. Contrary to expectations, FedAvg (0.945) outperforms FedProx (0.890) on AUPRC. We hypothesize that FedProx's proximal term, while stabilizing convergence, may over-regularize in settings where institutions have highly skewed fraud rates. FedProx does achieve slightly better F1 (0.207 vs 0.090), suggesting a different precision-recall trade-off.

Privacy at $\epsilon=5.0$ maintains viability. With $\epsilon=5.0$ (strong privacy), FedProx+DP achieves AUPRC of 0.692—still above the local-only average of 0.643. At $\epsilon=1.0$ (very strong privacy), performance degrades significantly (AUPRC 0.079), indicating that very aggressive noise injection overwhelms the model. Future work will explore adaptive differential privacy mechanisms and privacy-aware aggregation to improve this trade-off.

Communication efficiency. Total communication cost of 30 MB over 20 rounds across 15 institutions translates to approximately 100 KB per node per round. On a 3G connection (384 kbps), this takes under 3 seconds—making weekly federated training rounds feasible even in rural Uganda.

5. Discussion and Future Work

Our experiments confirm that federated learning can address the fundamental data scarcity challenge facing fraud detection in Uganda's microfinance sector. The 47% AUPRC improvement over local-only training, combined with near-perfect recall, demonstrates that collaborative model training yields practically useful fraud detection that no single institution could achieve alone.

Limitations. The current experiments use synthetic data (PaySim) rather than real Ugandan SACCO data. While PaySim is modeled on African mobile money transactions, real SACCO loan data exhibits different feature distributions and fraud typologies (ghost loans, collusion rings) not captured in PaySim. The schema heterogeneity module has not yet been tested with real multi-system data. The low F1 scores suggest threshold calibration and precision-recall trade-off tuning will be important for deployment. The DP results at $\epsilon=1.0$ highlight the need for more sophisticated privacy mechanisms.

Ongoing and future work. We are pursuing: (1) A **pilot deployment** with 3–5 SACCOs through FraudShield Uganda's partnerships to validate on real data. (2) Integration with the **Tazama platform** [26], the Linux Foundation's open-source fraud monitoring system for emerging markets. (3) Exploration of **vertical FL** to incorporate credit reference bureau and mobile money provider data. (4) **Adaptive differential privacy** mechanisms that calibrate noise to each institution's data characteristics. (5) Development of a **MoMTSim-FL** benchmark extending Azamuke et al.'s [14] Uganda-calibrated simulator with federated learning scenarios.

6. Conclusion

We introduced FedFraudShield, the first federated learning framework designed for fraud detection across Uganda's microfinance institutions and SACCOs. Born from the practical data challenges encountered by the FraudShield Uganda project, our framework demonstrates that federated learning achieves 47% higher AUPRC than institution-local models (0.945 vs 0.643) with near-perfect recall (0.991), while enabling institutions with no fraud history to gain detection capabilities. The framework maintains compliance with Uganda's DPPA 2019, with differential privacy at $\epsilon=5.0$ preserving performance above local baselines. The communication-efficient design (100 KB per node per round) makes deployment feasible even in low-bandwidth rural settings. As Uganda's SACCO sector continues its digital transformation, privacy-preserving collaborative approaches like FedFraudShield offer a path toward effective fraud prevention that no institution could achieve alone.

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